Embedded Systems Development

Exercise 4

1. Design an "auto light" using ATmega328P microcontroller and give your design as a schematic diagram.

   The "auto light" is an embedded system that automatically turns an LED on or off depending on the ambient light intensity. It measures the ambient light intensity using a transducer and automatically turns the LED on at nightfall and turns it off in the morning light.

   The "auto light" demonstrates how a physical phenomenon is sensed and converted into an analog voltage using a transducer, how the analog voltage is mapped to a discrete number and how a microcontroller processes this number to generate a digital output to control an actuator.

   Schematic Diagram

2. Write the required firmware for the "auto light" using the Arduino wrapper functions.

3. Write the required firmware for the "auto light" in C.

4. Extend the "auto light" design given in Question 1 to control a light connected to the domestic mains supply (230 V AC). Provide your design as a schematic diagram.



Warning - This extension involves working with 230 V AC mains voltage, which can cause serious injury, electric shock, fire or death. **Do not construct, test or modify a mains voltage circuit as part of this exercise unless it is carried out under the supervision of a suitably qualified person and in accordance with applicable electrical safety regulations. Do not connect mains voltage circuits to a breadboard or use exposed wiring. Appropriate electrical isolation, protection, insulation, enclosure and correctly rated components must be used**.

Schematic Diagram